Project Report: ESP32-Based Voltage and Current Protection System

Introduction

This project uses an ESP32 microcontroller to monitor voltage and current and protect an electrical load from overvoltage and overcurrent conditions. An LCD displays the measured values and system status, while a relay disconnects the load when a fault occurs. Three LEDs indicate normal operation, faults, and cooldown.

Objectives

Monitor voltage and current using sensor inputs.Disconnect the load when voltage or current exceeds the configured limit.Provide adjustable protection settings.Display readings and status on a 16×2 I²C LCD.Automatically reconnect the load after a safe cooldown period.Store settings in ESP32 non-volatile memory.

Hardware Components

ESP32,Relay module,Green LED,Red LED,Yellow LED,Three push buttons,LCD Display

Pin Configuration

ESP32 GPIO,GPIO 35,GPIO 33,GPIO 32,GPIO 14,GPIO 18,GPIO 19,GPIO 21,

Working Principle

The ESP32 reads voltage and current sensor signals .The measured values are filtered to reduce fluctuations .The controller compares the readings with the configured limits.If either limit is exceeded continuously for the trip delay, the relay disconnects the load.The red LED flashes during a fault.When the readings return to safe levels, the system enters cooldown and the yellow LED turns on.After 10 seconds of safe conditions, the relay reconnects the load and the green LED turns on.

Software Features

The program is developed using the Arduino framework for ESP32 and includes: LCD monitoring , Overvoltage and overcurrent protection , Adjustable trip and reconnection delays , Push-button settings menu , Preferences-based non-volatile settings storage,Serial commands for configuration and status , Three LED status indicators , Automatic fault recovery and relay control

Limitation : The ESP32-based voltage and current protection system provides a simple method of monitoring electrical parameters and disconnecting a load when configured limits are exceeded. The LCD, LEDs, push buttons, and automatic reconnection feature make the system easier to monitor and operate.